Greenkeeping

Science

May, 2026

Greenkeeping (A08582)

Short Title:	Greenkeeping
Faculty	Science and Computing
Department	Science
Cluster	Horticulture
Author	GMCMAHON
Credits	5

Level: Intermediate

Description of Module / Aims

This module aims to enable students to develop an understanding of the maintenance, management and construction of modern golf courses in an environmentally sound manner.

Programmes

HORT-0019	BSc in Horticulture (WD_SHORT_D)	3 / 6 / E
HORT-0019	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / E

Indicative Content

- Golf course architecture, design and planning
- Construction of golf course features
- Maintenance of golf course features
- Golf course ecology and habitats
- Impact of the rules of golf on course maintenance
- Golf course nutrition, growth regulation and modification of growing environments

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explore the principles of planning, construction and maintenance of golf courses.
2. Complete operations in golf course maintenance.
3. Explore the principles of environmentally sound management.
4. Examine the value of ecology and habitats and their impact on the golf course environment.
5. Relate the influence of the rules of golf on management and maintenance.
6. Justify the use of fertilisers and chemicals in golf course management.
7. Determine appropriate maintenance regimes for golf course features.
8. Examine current management practices on a golf course.

Learning and Teaching Methods

- Lectures
- Practical assignments.
- Reports

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5,6,7
Continuous Assessment	50%	
Assignment	25%	8
Practical	25%	2

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Beard, J. _Turf Management for Golf Courses_. 2nd ed. New Jersey: Wiley, 2001.

- Christians, N. and M.L. Agnew. _The Mathematics of Turfgrass Maintenance_. 4th ed. New Jersey: Wiley, 2008.
- Fagerness, M. and R. Johns. _Turfgrass Chemicals and Pesticides. A Practitioner'__s Guide_. 1st ed. New York: McGraw-Hill, 2003.
- Hayes, P. and R.D.C. Evans. _The Care of the Golf Course_. Bingley, West Yorkshire, England: The Sports Turf Research Institute, 1992.
- Hurdzan, M. _Golf Course Architecture: Evolutions in Design, Construction_ _and Restoration_. 2nd ed. New Jersey: Wiley, 2012.
- Lilly, S. _Golf Course Tree Management_. 1st ed. New Jersey: Wiley, 1999.
- Pira, E. _A guide to golf course irrigation system design and drainage_. 1st ed. New Jersey: Wiley, 1997.
- Vargas, J.M. and A.J. Turgeon. _Poa annua, Physiology, Culture and Control of Annual Bluegrass_. 1st ed. New Jersey: Wiley, 2003.

Requested Resources

- Lecture Room: Loose Seated